

Mobility Management and MANETs in Modern Wireless Communication Systems

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Introduction

Mobility management and Mobile Ad Hoc Networks (MANETs) play a vital role in modern wireless communication systems by enabling seamless connectivity and communication in dynamic environments.

Mobility Management

Mobility management ensures continuous network connectivity for moving users. It includes location management, handoff strategies, and Quality of Service (QoS). Location management tracks the position of mobile devices, while handoff strategies maintain active connections when devices move between networks. QoS ensures acceptable performance in terms of latency, bandwidth, and reliability.

Mobile Ad Hoc Networks (MANETs)

MANETs are infrastructure-less networks where nodes are mobile and self-configuring. Key characteristics include dynamic topology, decentralized control, and multi-hop communication. Major challenges include routing complexity, limited resources, security threats, and scalability issues.

Emerging Technologies

5G networks provide low latency, high data rates, and better security. Edge computing reduces delay by processing data closer to users. Future 6G networks are expected to use AI-driven mobility management, improve scalability, and support seamless global connectivity with advanced security mechanisms.

Conclusion

Together, mobility management and MANETs enable flexible and reliable wireless communication. Emerging technologies such as 5G, edge computing, and 6G are expected to overcome current challenges and enhance future wireless networks.